

JUSTIN CHEE

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Education

University of Bristol

MEng Aerospace Engineering

Bristol, UK

2025-2029

- **Relevant Modules Completed:** Fluids Mechanics, Thermodynamics, Materials
- **First Year Average:** 72.8% (First Class)

St. Joseph's Institution (International)

International Baccalaureate (IBDP)

Singapore

2021-2022

- **Higher Level:** Physics, Math, Chemistry (7,7,7)
- **Full Merit-Based Scholarship Recipient**

Professional Experience

Singapore Armed Forces

Feb 2023-May 2025

Lieutenant – Operations and Administration

- Managed HR administration for 1100 personnel, covering manpower, personnel administration, pay and benefits and workplace investigations; consistently achieved 95-100% compliance across 12 consecutive monthly audits.
- Spearheaded the administrative implementation of a recruit recovery programme, coordinating across army-wide HR, Medical and operational branches, recovering 90% of injured recruits to training.
- Led operation planning and execution of large-scale military events involving approximately 4000 participants and 16000 guests, developing contingency plans and coordinating with cross-functional stakeholders.

Singapore Youth Flying Club

Jan 2021-Sept 2021

Student Pilot

- Completed 22 hours as Pilot in Command and achieved a First Solo Flight, while developing a strong level of situational awareness and decision making.
- Applied principles of aerodynamics, navigation and aviation safety throughout training sorties.

Engineering and Technical Projects

CFD Analysis of an F1 Wing

Personal Project | Autodesk Fusion 360 | ANSYS Fluent

- Modelled a 1:1 scale multi-element F1 front wing and developed a CFD workflow (mesh generation, boundary-condition, solver configuration) across a 25° angle sweep, to quantify downforce and observe aerodynamic stall.
- Performed mesh analysis for solution reliability; diagnosed and resolved mesh generation failures through inflation layer optimization and proximity-based sizing, achieving a converged solution with a 1M computational budget.

Drone PID Build

Personal Project | MATLAB | Simulink

- Independently designed, tuned and validated a PID-controlled quadcopter attitude simulation, achieving a 13.49% overshoot and 4.45s settling time, while successfully recovering from disturbance rejection under closed-loop testing.

Mathematical Route Optimisation Using Bezier Curves

Nov 2021-May 2022

IB Mathematics Extended Essay | Mathematical Modelling

- Investigated the use of Bezier Curves for mathematical modelling and route optimization, modelling paths through controlled points. Used Desmos to model, visualise and compare Bezier Curve based route optimisation with Polynomial Interpolation, demonstrating that Bezier Curves yielded a more optimised route with a lower curvature.

DIY Wind Tunnel and Aerofoil investigation

Jan 2022-May 2022

IB Physics Internal Assessment | Experimental Aerodynamics

- Designed and constructed a simple wind tunnel to determine the effect of varying freestream velocities on an aerofoil. Conducted controlled experiments to analyse and evaluate the relationship between airspeed and lift forces.

Skills and Interests

Technical Skills: CFD Analysis (ANSYS), CAD (Fusion 360), Python, MATLAB

Professional Skills: High Pressure Decision Making, Cross-Functional Collaboration, Events and Operations Management, Situational Awareness, Leadership and Team Management, Problem Solving

Interests: Athletics, Football, Cooking, Gaming, Aviation